STATISTICAL ARBITRAGE ON INDIAN LISTED EQUITIES

Distance / Minimum SSD vs Engle-Granger Pairs Trading

Quantitative Researcher Case Assignment - Final Research Memo

Objective

Test whether a simple relative-value pairs strategy on Indian equities survives point-in-time universe construction, multiple-testing control, realistic short-leg constraints, transaction costs and a reserved final evaluation block. I compare the assignment benchmark - distance/minimum SSD - with the Engle-Granger (EG) two-step method on exactly the same NIFTY Pharma universe, calendar and cost assumptions.

Headline finding. Under the frozen 12-month formation / 6-month trading specification, EG clearly outperformed SSD, but neither result supports a production-ready edge.

EG returned +4.34% net versus -10.38% for SSD, with a 70% versus 40% convergence rate and a much smaller drawdown. However, EG made only 10 trades, made no OOS trades, and changing the formation-window length reversed the method ranking.

Primary result (1x costs)	SSD	Engle-Granger
Net cumulative return	-10.38%	4.34%
Trades	20	10
Convergence rate	40.0%	70.0%
Maximum drawdown	-12.90%	-2.70%
NIFTY 500 beta	0.008	0.001

What I believe

The primary evidence favors EG because its stricter relationship test produced fewer but higher-quality trades. SSD was already unprofitable before costs, so its failure is primarily a pair-selection problem rather than an India-cost problem. My confidence is moderate-to-low because the sample is small and formation-window sensitivity is large.

Research discipline

The final comparison is made on 1-Feb-2017 to 31-Jul-2026 with identical 0x/0.5x/1x/2x cost grids. Both final method audits and the comparison audit pass. Negative and inconclusive results are retained rather than tuned away.

1. Universe and Data

Universe. I restricted the search ex ante to NIFTY Pharma and reconstructed historical membership rather than applying today's constituents backwards. This deliberately conditions pairs within one industry, reduces the candidate search space and avoids obvious cross-sector matches with weak economic justification. The trade-off is that opportunities outside pharmaceuticals are excluded. I did not run a pharma-only versus cross-sector comparison, so I cannot tell whether sector conditioning improved pair quality or merely reduced the opportunity set; I treat that as a limitation.

Data / investability item	Final treatment
History	2016 through 31-Jul-2026; 2,615 NSE sessions; trading comparison begins 1-Feb-2017
Point-in-time membership	Historical NIFTY Pharma membership; 36,646 member-price rows
Corporate actions	446 events reviewed; dividends in total return; splits/bonuses/rights adjusted; demergers treated as structural breaks
Missing data	No forward-fill; 14 legitimate member-date gaps retained
Formation quality	>=95% coverage; max missing streak <=5; one structural segment; price >= ₹50
Liquidity	Median daily traded value >= ₹10 crore over formation window
Shortability	NSE single-stock future required; entry-date futures availability checked for actual short leg
Formation dates	19 rolling formation dates; minimum 8 investable names; 230 investable stock observations

Data lineage and corporate actions

NSE bhavcopy is the primary price source. Raw traded fields are preserved separately from the cleaning layer. The total-return series reinvests dividends for the long-side economic series, mechanically adjusts splits and bonuses, applies a TERP-style rights adjustment, and does not stitch across demerger structural breaks. ISIN and symbol-transition audits were used to avoid silently joining different companies. Muhurat sessions were explicitly restored and exchange holidays were distinguished from per-stock suspensions.

Tradability

The formation universe is the intersection of historical index membership, data-quality rules, the ₹10 crore traded-value threshold and historical stock-futures eligibility. The final primary trade ledgers contain 20 SSD trades and 10 EG trades; all were executable under the stock-futures availability test. The short leg is therefore treated as a listed single-stock future rather than an overnight naked cash short.

Why this universe is defensible

- Point-in-time membership addresses survivorship bias directly.
- Pharma-only conditioning provides an economic prior and sharply reduces the false-discovery problem.
- Liquidity, price and F&O filters trade breadth for a more realistic implementation.
- The remaining realism gaps - actual futures basis/roll, integer lot sizes, ASM/GSM and locked circuits - are disclosed rather than silently ignored.

2. Methodology

Common protocol. Both methods use a rolling 12-month formation window followed by a 6-month trading window. Pair choice, spread definition and formation statistics are frozen during trading. A signal observed at the close is executed on the next common market observation. No suspended-price forward-fill is allowed. Up to four non-overlapping pairs may be active, with a 25% gross portfolio cap per pair; unused slots remain in cash at 0%. The primary specification uses no separate stop-loss: unresolved positions are force-closed at the six-month window end. A 4-standard-deviation stop was tested only as a robustness check, not added after seeing the primary results.

2.1 Distance / minimum SSD

Each formation-window total-return series is normalized to the same starting value, following the distance-pairs idea of Gatev et al. (2006). For every candidate pair, I sum the squared day-by-day differences between the two normalized paths; smaller SSD means the paths were historically closer. Across the 19 formation dates, SSD examined 1,362 candidate pair-period relationships. To control the risk of selecting a pair that only looked close by chance, I split the 12-month formation history into two halves: a pair must rank in the closest 10% in the first six months and again in the closest 10% in the second six months. Twenty-five pair-periods survived this two-half validation and 19 non-overlapping pair-periods were finally selected.

Trading rule: enter when the frozen normalized spread moves beyond +/-2 formation standard deviations; buy the cheap leg and short the rich leg. Exit when the spread crosses the formation mean, otherwise force-close at the six-month window end. Legs are equal-capital: 12.5% long and 12.5% short for a full 25% pair slot.

2.2 Engle-Granger two-step

For every unordered pair I apply the Engle-Granger two-step idea (Engle & Granger, 1987) in both regression directions. The relationship coefficient and residual gap are estimated on formation data only, and the unit-root test uses Engle-Granger/MacKinnon values rather than ordinary ADF values. The same 1,362 unordered pair-period relationships therefore generate 2,724 directional tests. Both directions count when controlling false discoveries; Benjamini-Hochberg correction is applied across all directional tests on each formation date. The primary lag rule is AIC, with BIC and fixed lag 1 as sensitivity checks, and a positive relationship coefficient is required. Twelve directions survive the correction, corresponding to six final unordered pair-periods.

Trading rule: enter at +/-2 frozen residual standard deviations and exit on a return through the frozen residual mean, with the same next-observation execution and six-month forced close. Within a 25% pair slot, A:B absolute notionals are scaled in proportion to 1:beta.

2.3 Costs and implementation

Item	Assumption
Cash leg	30 bps round trip
Single-stock future	8 bps round trip
Cost grid	0x gross; 0.5x; 1x primary; 2x
Futures financing	20% of short-futures notional funded at 8% p.a. (model assumption)
Capacity	Each execution leg capped at 1% of that day's traded value

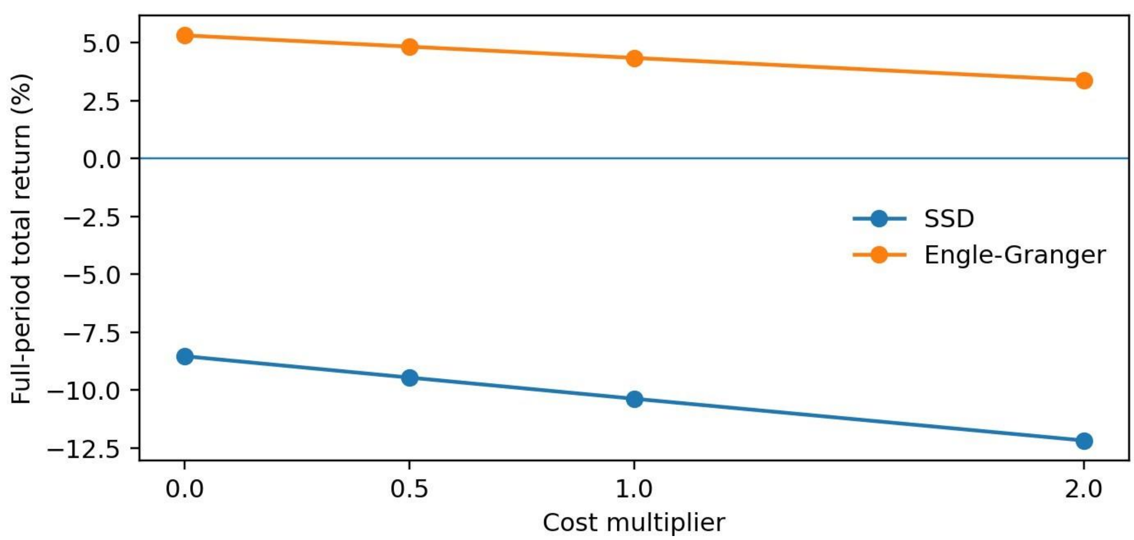
3. Main Results

Metric	SSD	Engle-Granger
Selected pair-periods	19	6
Trades	20	10
Hit rate	50%	70%
Convergence / forced close	40% / 60%	70% / 30%
Gross return	-8.54%	5.32%
Net return (1x)	-10.38%	4.34%
CAGR	-1.17%	0.46%
Sharpe	-0.39	0.24
Max drawdown	-12.90%	-2.70%
Turnover / NAV	10.11	5.17
Capacity	₹0.87 crore	₹1.48 crore

Interpretation

EG dominates the primary specification on both return and trade quality. Crucially, SSD loses -8.54% even before costs, while EG earns +5.32% gross. Therefore the result is not simply that India trading costs killed an otherwise profitable SSD signal. EG also trades half as often, turns over roughly half as much capital, and reaches the expected convergence exit much more often.

Figure 1. Full-period return across identical cost assumptions



At every cost level EG remains positive and SSD remains negative. At the 1x primary assumptions, costs consume about 1.84% of initial NAV for SSD versus 0.95% for EG, reflecting SSD's higher trading activity and longer average holding period.

4. Trade Quality, Attribution and OOS

Trade quality explains more of the method difference than costs. SSD produced 20 trades, only 8 of which returned to the formation mean; 12 were force-closed. EG produced 10 trades, 7 converged and 3 were force-closed. Mean / median holding periods were 94.7 / 89.5 days for SSD and 78.2 / 70.5 days for EG. Mean / median net portfolio contribution per trade was -0.52% / -0.14% for SSD versus +0.43% / +0.94% for EG. The holding-period distribution below is supplied from the final ledger; complete method-specific monthly-return tables and daily NAV/exposure histories accompany the submission as CSV outputs.

Holding period	SSD trades	Engle-Granger trades
0-30 days	4	2
31-60 days	3	2
61-90 days	3	2
91-120 days	3	2
121-180 days	6	2
>180 days	1	0

Holding-period counts sum to 20 SSD trades and 10 EG trades. Full monthly returns and daily exposure series are supplied in companion result files.

Loss concentration was meaningful but not hidden. SSD's two largest losing pair groups were BIOCON__GLENMARK (-5.22%) and DRREDDY__ZYDUS (-4.94%). EG's largest losing pair was DIVISLAB__TORNTPHARM (-2.09%). The top 10% of trades contributed about 33.9% of SSD positive P&L and 18.7% of EG positive P&L, so neither result is solely one lucky winning trade.

Market exposure

Metric	SSD	Engle-Granger
NIFTY 500 beta	0.008	0.001
NIFTY 500 correlation	4.6%	1.0%
Average gross exposure	13.8%	6.0%
Average absolute net exposure	0.44%	0.82%

Both strategies have near-zero realised beta to the NIFTY 500. The relative result therefore is not explained by a disguised directional equity-market bet.

Reserved final block

OOS / final block	SSD	Engle-Granger
Selected pair-periods	6	0
Trades	9	0
Net return at 1x	+0.09%	0.00% (cash)
Interpretation	Essentially flat	No OOS trading evidence

I do not interpret EG's zero OOS return as success: it found no qualifying pairs and therefore generated no trading evidence. SSD did trade but finished essentially flat after costs. In addition, earlier implementation diagnostics exposed some information from the reserved block before the final asymmetry/validation specifications were frozen. The final block is therefore described as constrained pseudo-OOS rather than a perfectly untouched holdout.

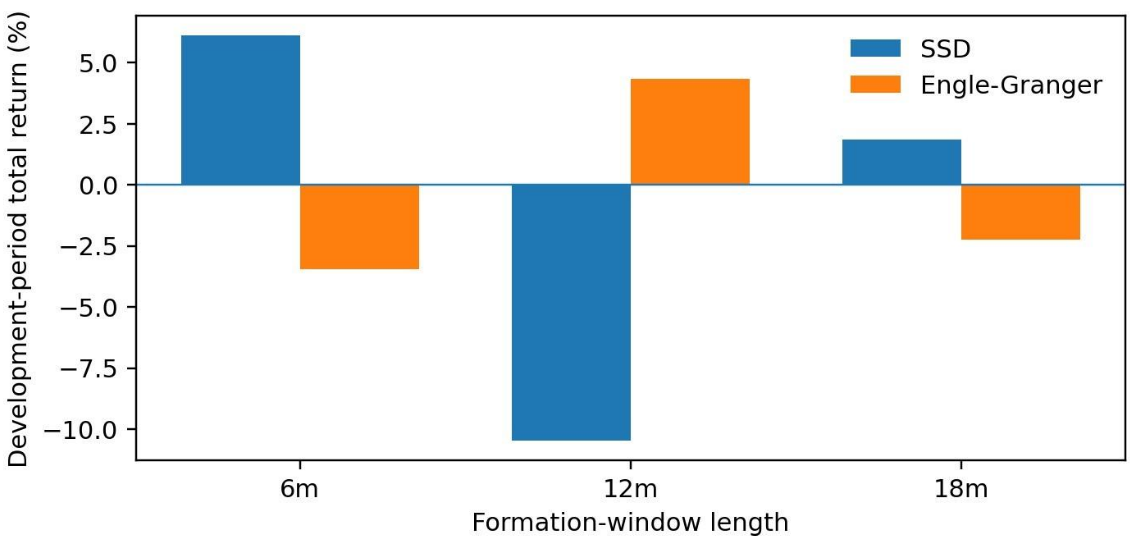
5. Robustness

Threshold changes do not rescue SSD and do not overturn EG under the primary 12-month formation design. The main weakness is instead formation-window dependence: changing the amount of history used to form pairs reverses the ranking between methods.

Development-period variant	SSD	Engle-Granger
Entry 1.5 SD	-10.84%	+5.87%
Primary: entry 2 SD	-10.46%	+4.34%
Entry 2.5 SD	-10.61%	+6.86%
Alternative exit	-10.43%	+2.24%
Formation 6 months	+6.10%	-3.45%
Formation 18 months	+1.85%	-2.24%
Top 75% liquid universe	+2.56%	+3.92%
Half-life 5-60 sessions	-8.32%	+1.51%*

*The EG half-life result contains only one selected pair and one trade, so it is not strong evidence.

Figure 2. Formation-window sensitivity reverses the method ranking



What this changes

The correct conclusion is not "EG is universally better." The defensible claim is narrower: EG materially outperformed SSD under the pre-specified 12-month formation / 6-month trading specification. A six- or eighteen-month formation window makes SSD positive and EG negative. This specification sensitivity is the strongest reason not to treat the +4.34% EG result as a production-ready edge.

6. Indian-Market Realism and Limitations

Short leg. Multi-day cash shorting is not assumed. The actual short stock must have an NSE single-stock future at entry. No stock-borrow fee is charged because the short is implemented through futures. Instead, I explicitly charge funding on assumed futures collateral. The funding assumption is 20% of futures notional at 8% per year; this is a model assumption rather than an observed historical margin series.

What is not fully reconstructed

- Actual historical futures basis and settlement-price paths. Futures P&L is approximated with a corporate-action-adjusted spot proxy that excludes shareholder cash dividends on the short proxy.
- Exact monthly futures contract rolling and roll gains/losses.
- Integer NSE futures lot sizes; the backtest sizes continuous notionals, so small portfolios may not reproduce the theoretical hedge exactly.
- Date-by-date ASM/GSM surveillance state and locked-circuit execution. A residual risk remains that a theoretical EOD fill was unavailable in practice.
- Perfectly untouched OOS. The final block was not used to tune thresholds or costs, but earlier diagnostics exposed some information; I therefore label it constrained pseudo-OOS.

Capacity and settlement

I cap each execution leg at 1% of that stock's actual daily traded value. The resulting conservative bottleneck capacity is about ₹0.87 crore for SSD and ₹1.48 crore for EG. Cash settlement is T+1; this mainly affects capital tie-up rather than the EOD signal, but live implementation would need explicit cash and margin scheduling.

Assumptions most likely to be wrong

- The 12-month formation window is structurally stable enough to identify relationships that persist for the next six months.
- The simplified futures proxy plus 20% collateral / 8% funding assumption is close enough to realized futures basis, roll and margin economics.
- The F&O/liquidity filters are sufficient to approximate actual executability despite unmodelled circuits and surveillance restrictions.

7. Conclusion, Confidence and Next Work

Conclusion. Under the frozen primary specification, EG is the stronger of the two methods. It returns +4.34% net versus -10.38% for SSD, trades less, converges more often and experiences a much smaller drawdown. The fact that SSD loses before costs is especially informative: historical path similarity alone did not identify reliable future convergence. EG's stricter relationship test appears to improve pair quality.

Confidence. I would not deploy either strategy from this evidence. EG has only ten trades and no actual OOS trades; SSD is negative in development and only flat in the final block. Most importantly, formation-window sensitivity reverses the method ranking. I therefore view the result as evidence that EG is a better filter under this particular design, not evidence of a persistent standalone alpha.

What would change my mind

Confidence would rise materially if the same ranking survived a truly untouched future period, actual futures-chain implementation, and a wider pre-registered universe without changing the formation rule. Confidence would fall further if the EG advantage disappeared once real basis/roll/lot-size effects were included or if a fresh OOS block again produced no qualifying trades.

What I would do with four more weeks

- Reconstruct actual single-stock futures settlement prices, contract rolls, margin schedules and integer lot sizing; then rerun both methods with the same execution engine.
- Rebuild historical ASM/GSM and circuit-lock states and reject fills that could not have occurred.
- Pre-register a broader but still sector-conditioned universe and validation protocol, then hold back a genuinely untouched forward period.
- Investigate structural-break diagnostics and whether pair relationships can be retired before the full six-month forced close without tuning on realised P&L.

Final assessment

Defensible result: EG beats SSD under the primary specification, but the evidence is too sparse and specification-sensitive to justify live deployment. The negative/flat findings are part of the result, not implementation failures.

References

Gatev, E., Goetzmann, W.N. & Rouwenhorst, K.G. (2006). Pairs Trading: Performance of a Relative-Value Arbitrage Rule. Review of Financial Studies, 19(3).

Engle, R.F. & Granger, C.W.J. (1987). Co-integration and Error Correction: Representation, Estimation, and Testing. Econometrica, 55(2).

Krauss, C. (2017). Statistical Arbitrage Pairs Trading Strategies: Review and Outlook. Journal of Economic Surveys, 31(2).

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Companion deliverables include FINAL_TRADE_LEDGER.csv, FINAL_METHOD_COMPARISON.csv, the reproducible code/archive and audit files. Method-specific result CSVs also provide the complete monthly return table, holding-period distribution, trade-statistics summary and daily NAV/gross/net-exposure history required by the backtest specification. Every graphical figure in this memo is reproduced by `python src/make_submission_exhibits.py`.